CBPCA IMPROVES A *HOME ENERGY* HOME

by ELKA KARL


CBPCA

Contractor Eric Sulter, whose company, Air Movement, is located in San Francisco, takes part in the CBPCA training.

As was described in the last issue of *Home Energy*, Carol Markell, the magazine's advertising and marketing manager, had volunteered her house for use during a training for the California Building Performance Contractors Association (CBPCA), a nonprofit organization that introduces contractors—and the general public—to home performance services (see "Training Day," *HE* July/August '05, p. 24). During the training, Markell's house was run through several diagnostic tests, which tested her home's performance while educating building contractors on the process of building performance testing.

About a month after the training was completed, CBPCA trainers Rick Chitwood and Tim Locke and students returned to Markell's Kensington, California, home to conduct a complete energy-feature retrofit. The home received a new heating system, a new duct system, a new water heater, a new gas dryer, warm floors in the bathroom, a quiet bathroom exhaust fan, infiltration reduction, high-efficiency air filtration, insulation in the ceiling and stemwalls, a lighting improvement, and a cleaned and sealed crawlspace (see "Test-Out Results").

Robert Mitchell and his crew were responsible for the job and the paperwork. It's hard to say whether having two instructors and 16 new home performance contractors at the job was a help or a hindrance, but Mitchell managed to complete the retrofit in the allotted three days.

"It was all done pretty much how Rick explained it to me," says Markell. "The crawlspace was [finished] much more completely than I realized [with an added vapor barrier]. They put a lock on the crawlspace access so that no one could get into it, like animals. The new gas dryer was put on a pedestal and the duct for the dryer is much improved. My clothes dry in under one hour; before, it could take three hours."

The old 80,000-Btu furnace was replaced with a 15,000-Btu/hr combined hydronic air handler that is run off the water heater. "Rick warned me that it may take longer to heat up the home in the morning due to the smaller furnace, but it actually heats up faster," Markell said.

The goal of the improvements was to enhance comfort and reduce energy bills. The predicted space heating reduction is 78% (from $406 per year to $90 per year) and the total bill reduction is predicted to be 48% (from $850 to $443 per year). To take full advantage of the improvements, a list of system operation recommendations was provided. CBPCA recommended that Markell keep the bedroom doors open as much as possible, so that air supplied to each bedroom can get back to the return grille in the hall. Opening interior doors also allows for better temperature communication between the bedrooms and the thermostat in the hall. When windows are opened in the summer, both of the thermostats should be switched off.

To remove moisture from the bathroom, CBPCA recommended that the bathroom exhaust fan always be used when showering. ("I have to remember to turn on the bathroom fan," admits Markell.) The twist timer should be set to run the fan after a shower for 15 minutes or more (except in the summer when the windows are open and the floor heat and furnace are off). One possible future improvement could be to replace the bathroom fan twist timer

Test-Out Results

A key component of the CBPCA home performance training is a completed work commissioning or "test-out." So for you more technical types, here are the numbers:*

Building Air Infiltration

Pre-test: 1,935 CFM_{50}
Sealing target: 900 CFM_{50}
Post-test: 824 CFM_{50} (41 CFM_{NAT} estimated, 0.29 ACH_{NAT} estimated)

Bathroom Mechanical Ventilation

Goal: 8 ACH in the bathroom

A 70-CFM Panasonic bathroom exhaust fan was selected. The first time the exhaust flow rate was measured the flow rate was 63 CFM. The exhaust duct was rerouted and shortened and the final flow was 77 CFM. This provides 11 ACH for the bathroom and 0.55 ACH for the whole house. The fan is controlled by a 30-minute twist timer.

Duct Leakage

Pre-test: 395 CFM_{25}
Sealing target: 25 CFM_{25}
Post-test: 11 CFM_{25}

Crawlspace Ventilation

The crawlspace ventilation fan can depressurize the crawlspace 1.3 Pa.

Room Air Flow and Static Pressures

Room Pressures	Design (CFM)	Measured Flow (CFM)	Deviation (%)
Living N/A	241	283	+17
Dining N/A	178	205	+15
Bedroom + 3.5 Pa	123	137	+11
M. Bedroom + 3.8 Pa	113	113	0
Total Flow	**655**	**738**	**+13**
Return static	- 0.204 in		
Supply static	+0.139 in		
Total Static	**-0.343 in (design static 0.35 in)**		

The manufacturer's fan curve shows that there should be 658 CFM at 0.34 in of static. The actual air flow was 738 CFM (13% higher than the fan curve). Air flow will decline as the air filter gets dirty, the fan blades get dirty, and the coil fouls.

Water Heater Combustion

Steady state efficiency: 79.8%
Carbon monoxide: 15 ppm

* results provided by Rick Chitwood

control with a control system that could cycle on the fan to provide whole-house ventilation, humidity control, or carbon-dioxide-level control if further improvement in the indoor air quality were desired.

The fireplace damper should be kept closed except when the fireplace is used. Phantom loads from the audio/video equipment in the living room and the computer equipment in the shed could be eliminated by placing them on a switched plug strip that is switched off between uses.

Markell had computer equipment in the shed; CBPCA recommended that she move the equipment into the house. Heating the shed with an electric heater is 2.6 times as expensive as the gas heat in the house. In addition, kitchen ventilation could be improved by installing a range hood when Markell remodels the kitchen.

For the next year, CBPCA asked Markell to mail a copy of her monthly utility bill to Rick Chitwood at CBPCA. This will ensure that any problems with the heating system or system operation are identified in a timely manner.

To maintain the new heating system, CBPCA recommends that Markell replace the 24 inch x 24 inch x 2 inch air filter at the return air grille once per year in the fall. When she is changing the filter, it is recommended that she also vacuum out the supply boots. The crawlspace should be checked once per year for signs of moisture, infestations, or odor. The crawlspace should be kept closed and locked at all times to prevent damage to the duct system, insulation, and vapor barrier. Finally, sediment should be flushed from the bottom of the water heater once per year and maintenance required by the water heater manufacturer should also be performed.

Markell has already noticed positive changes in the house. "The house temperature is much more even. The bathroom floor is wonderful. The house was not hot [in June] when it was hot out. But I'm interested in seeing what happens during a three- or four-day heat wave."

*Elka Karl is an associate editor at **Home Energy**.*

For more information:

For more information about CBPCA's training programs, go to www.cbpca.org.